SCIE 1120 Chemistry and Life

Origin of Life on Earth

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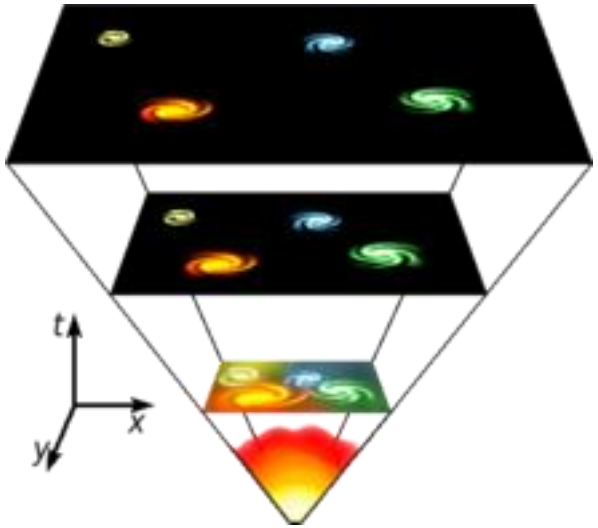
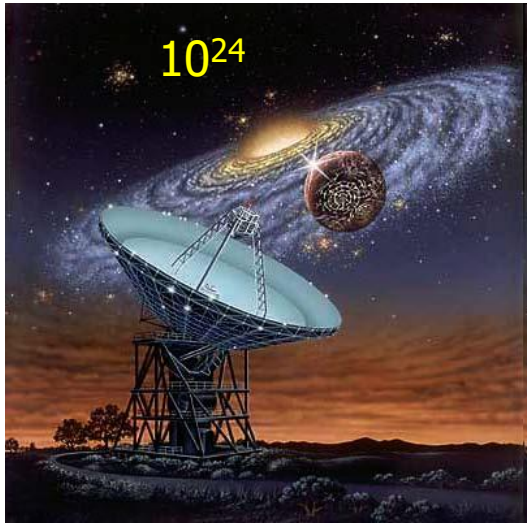
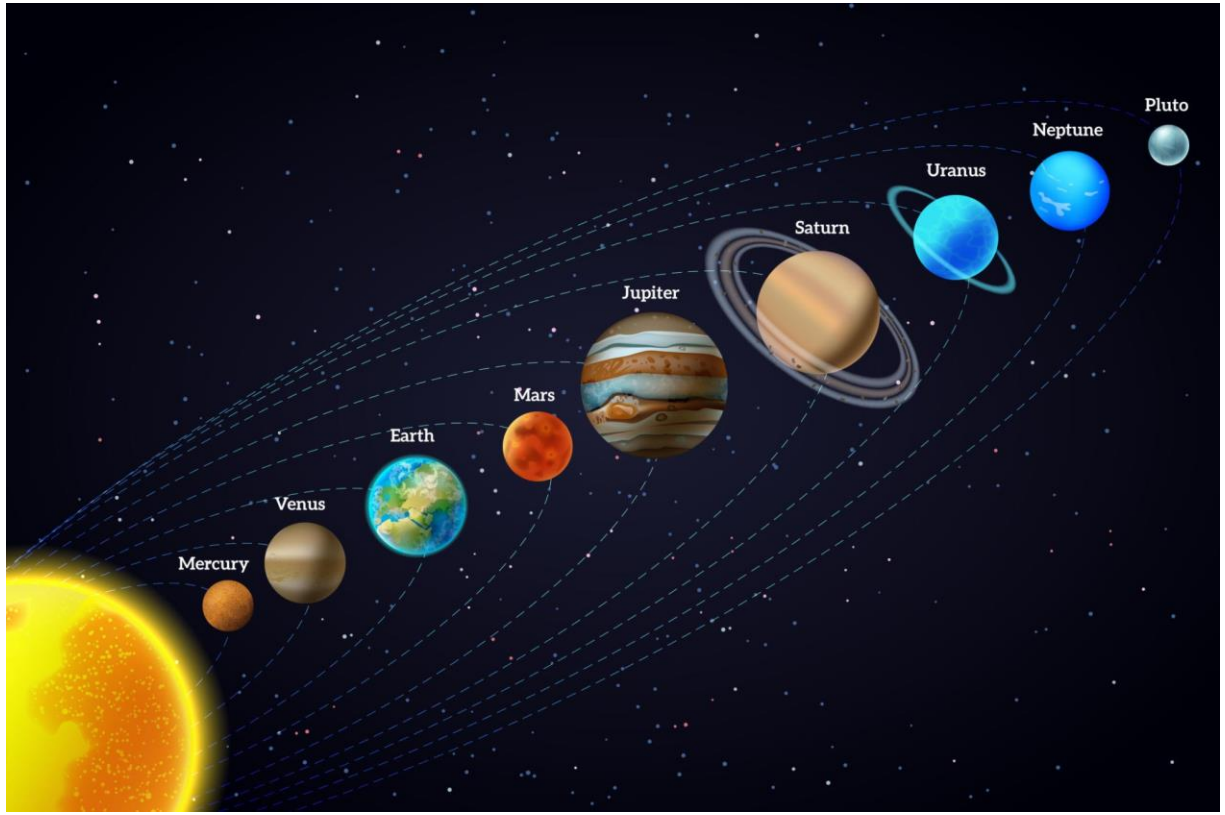
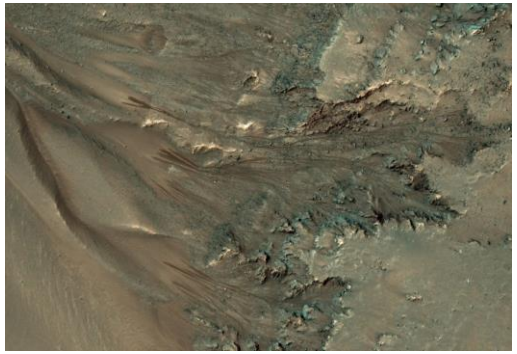
Life in the universe







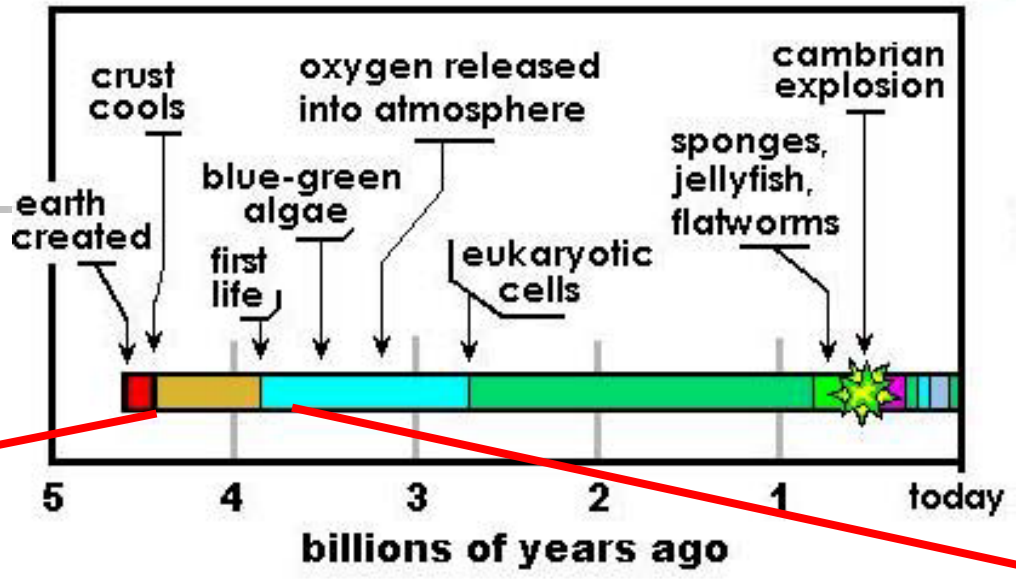
13.3 to 13.9 billion (10^9) years ago

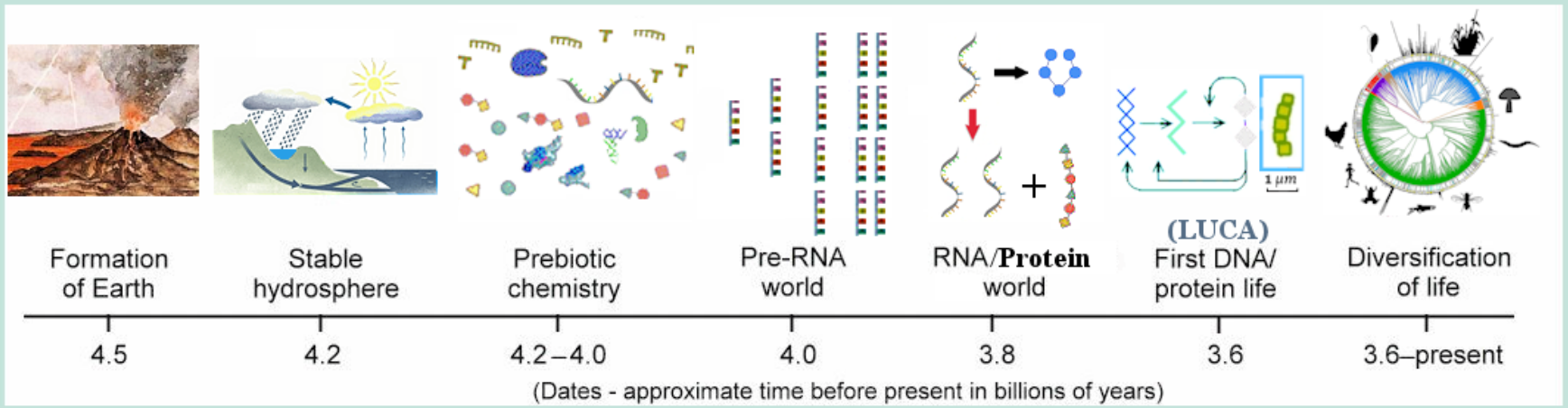


Mars has ice

There are plenty of planets in the universe, many are like our own.

Formation of Earth

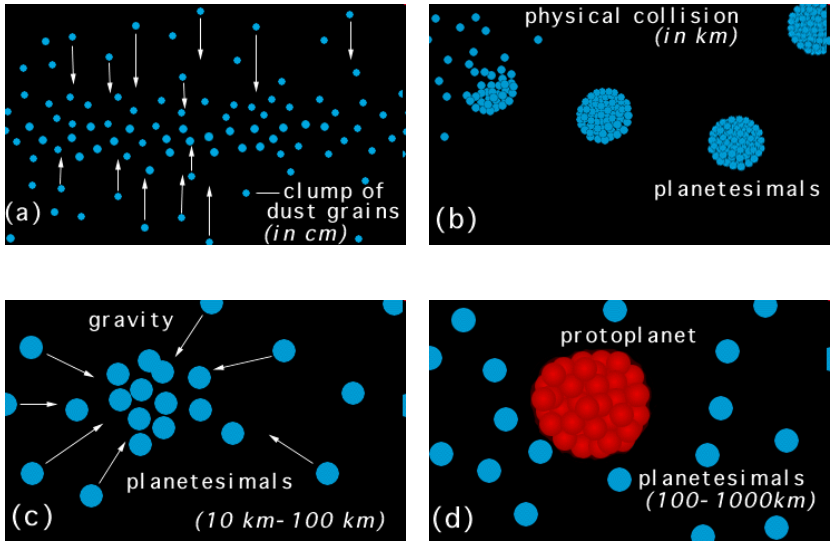


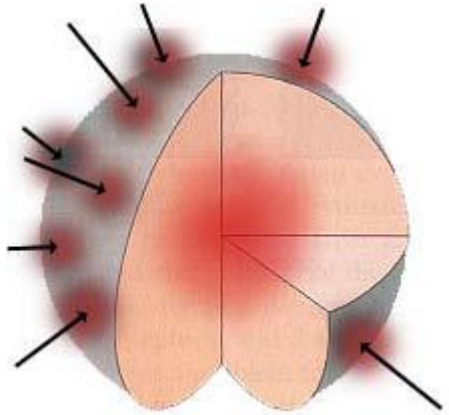


Timeline of events pertaining to the early history of life on Earth.

Ave temp 4000 160 41 35 29 C

The Earth formed over 4.5 Billion Years Ago



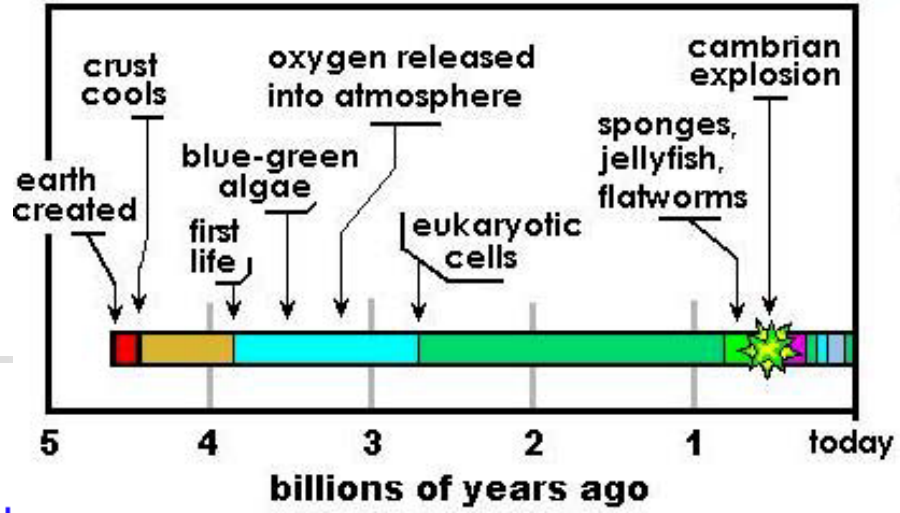


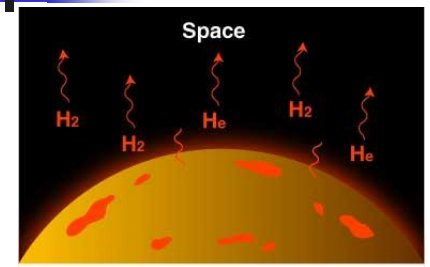
行星体是形成的一种天体,由尘土、岩石和其他物质构成。
这个词源于“无穷小”这一概念，
指的是小到看不见或无法测量的物体。
早期地球因放射性衰变、压缩和撞击而升温。随着时间的推移，**地球内部**的温度逐渐升高，接近铁的熔点，从而形成了地球磁场

- A **planetesimal** is an object formed from dust, rock, and other materials. The word has its roots in the concept infinitesimal, which indicates an object too small to see or measure.

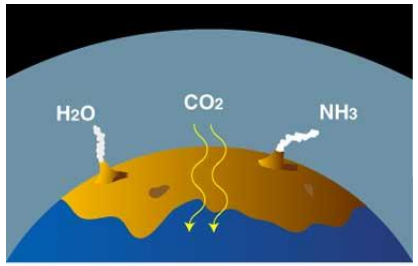
  Early Earth heats up due to **radioactive decay, compression, and impacts**. Over time the temperature of the planet interior rises towards the Fe melting point creating earth magnetic field.
- Volcano

Stable hydrosphere

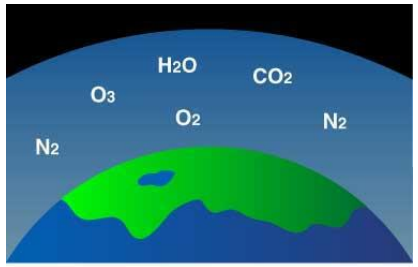




Atmosphere #1



Atmosphere #2



Atmosphere #3

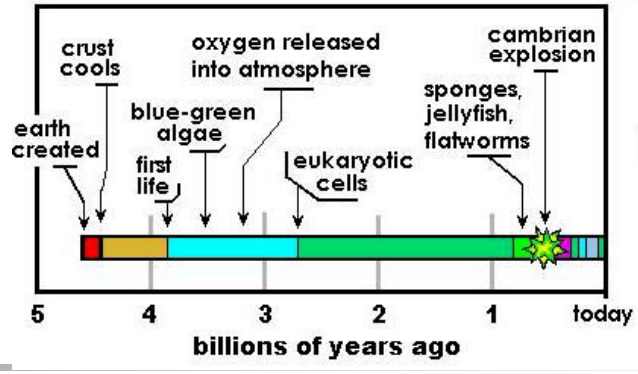
Evolution of Our Atmosphere

Original Atmosphere of H/He Lost

- Secondary Atmosphere liberated due to planetesimal collisions with proto-earth that released volatiles trapped in grains
- $CO_2 + H_2O$ (liquid) $\Rightarrow H_2CO_3$ (acid rain) carbonate acid
- $H_2CO_3 + Ca^{++} \Rightarrow CaCO_3$ (limestone)
- $H_2CO_3 + CaMg(CO_3)_2 \Rightarrow$ dolomites
- $N_2 \Rightarrow$ chemically inert so it stays in atmosphere; N_2 cycle develops later
- $O_2 \Rightarrow$ not present in original atmosphere; gotta make some critters first!

Delivery of water by icy planetesimals and comets: After condensation of water vapor produced the earth's oceans, thus sweeping out the carbon dioxide and locking it up into rocks, our atmosphere was mostly nitrogen and some CO2.

冰质星子和彗星带来的水：地球上的海洋形成于水蒸气凝结之后，**从而清除了大气中的二氧化碳，并将其封存在岩石中，**于是我们的大气层主要由氮气和少量二氧化碳构成



厌氧微生物利用硫酸盐而非氧气来满足其能量需求

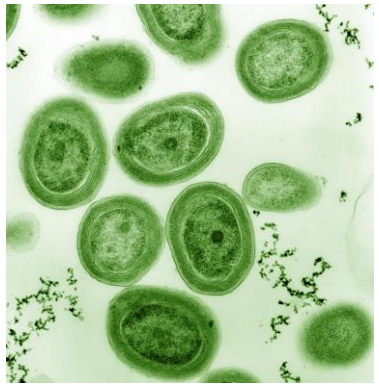
- SO_4^{2-} , anaerobic microorganisms used sulphates instead of oxygen to meet their energy needs (next to hydrothermal vent in the bottom of ocean)
- CO2+sun light, anaerobic microorganisms thrive with photosynthesis and release O2.

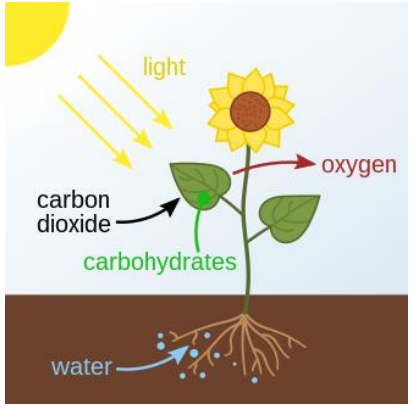
光合作用

anaerobic microorganisms：Any organism that does not require molecular oxygen for growth.

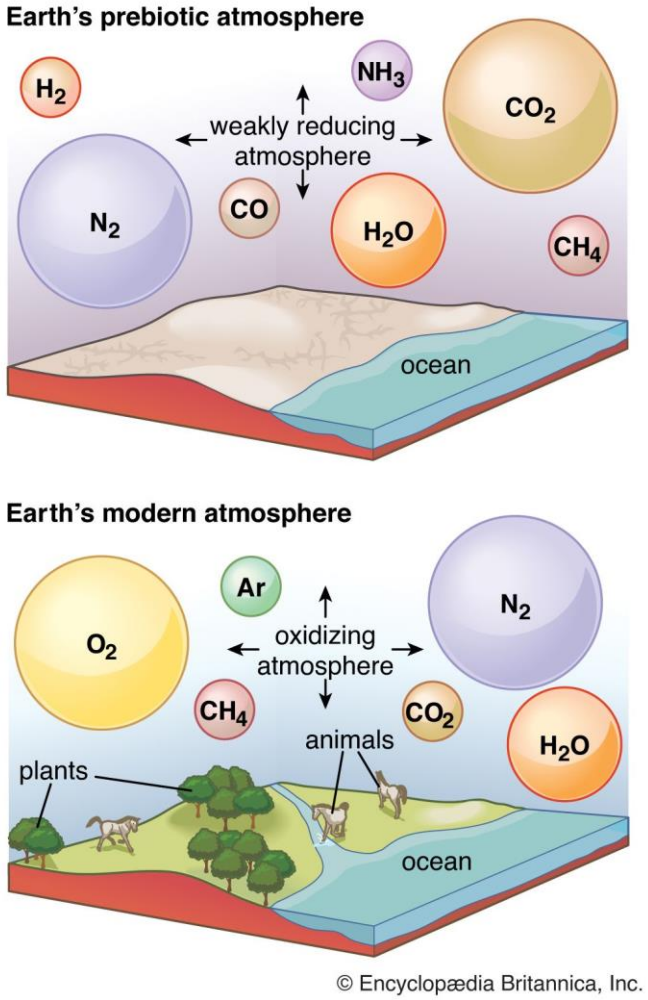
Organism:is any living system that functions as an individual entity 生物体

Living system: growth, energy transformation, reaction to stimuli, and reproduction.

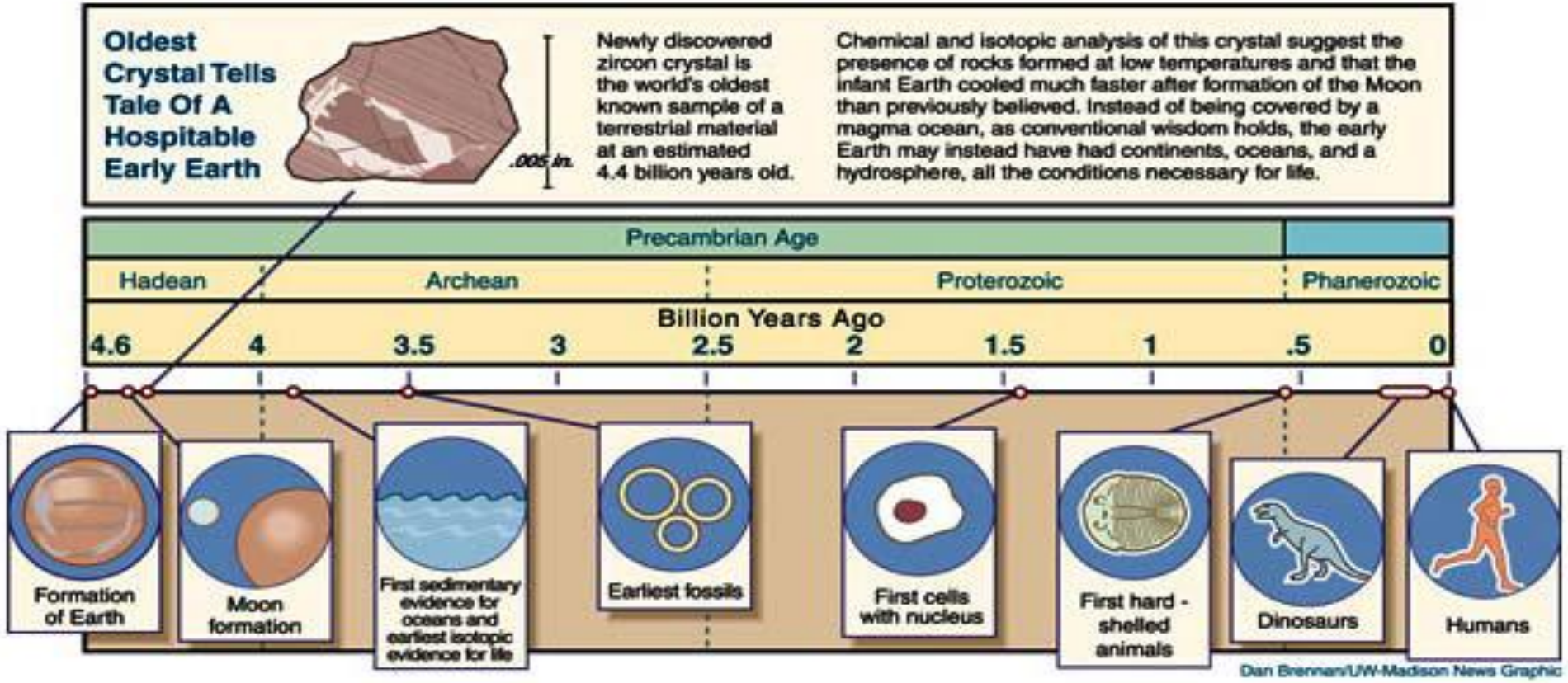




Cyanobacteria (2.4BYA) obtain energy via photosynthesis 蓝细菌；光合作用



Ave. oxygen concentration 21% in 1 BY 21%一个大气压



0.1-.3mm

锆石（硅酸锆晶体）

Tiny zircons (zirconium silicate crystals) found in ancient stream deposits indicate that Earth developed continents and water -- perhaps even oceans and environments in which microbial life could emerge -- 4.3 billion to 4.4 billion years ago, remarkably soon after our planet formed. The presence of water on the young Earth was confirmed when the zircons were analyzed for oxygen isotopes and the telltale signature of rocks that have been touched by water was found: an elevated ratio of oxygen-18 to oxygen-16. (because 18O is more abundant in ocean due to its mass)

Theories about the origin of life

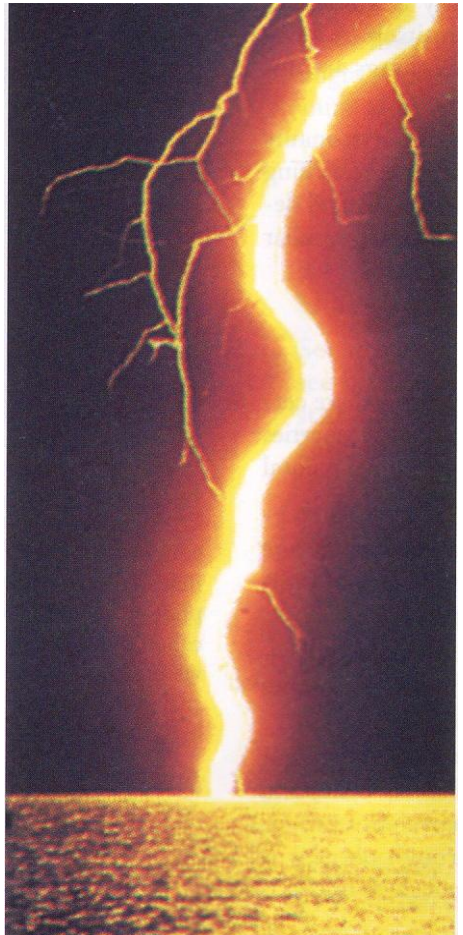
- **Special creation**: life forms may have been put on earth by supernatural or divine force.
- **Extraterrestrial origin**: 外星起源 life may not have originated on earth at all; instead, life may have infected earth from other planets

自发起源

Spontaneous origin:

life may have evolved from inanimate matter, as 无生命 association of molecules became more and more complex.

Theories about the origin of life



Lightning

Before life evolved, the simple molecules in the earth's atmosphere combined to form more complex molecules.

The energy that drove these chemical reactions may have come from lightning and forms of geothermal energy.

地热能

前生命化学是在原始地球环境可能存在的条件下，在水溶液中进行的有机化学反应，其目标是生成具有生物学意义的化合物。基础前生命化学使用简单且具有反应活性的有机化合物，例如氰化氢(HCN)、氰乙炔(HC3N)、甲醛(HCHO)或它们的低聚物。

前生命化学

Prebiotic Chemistry

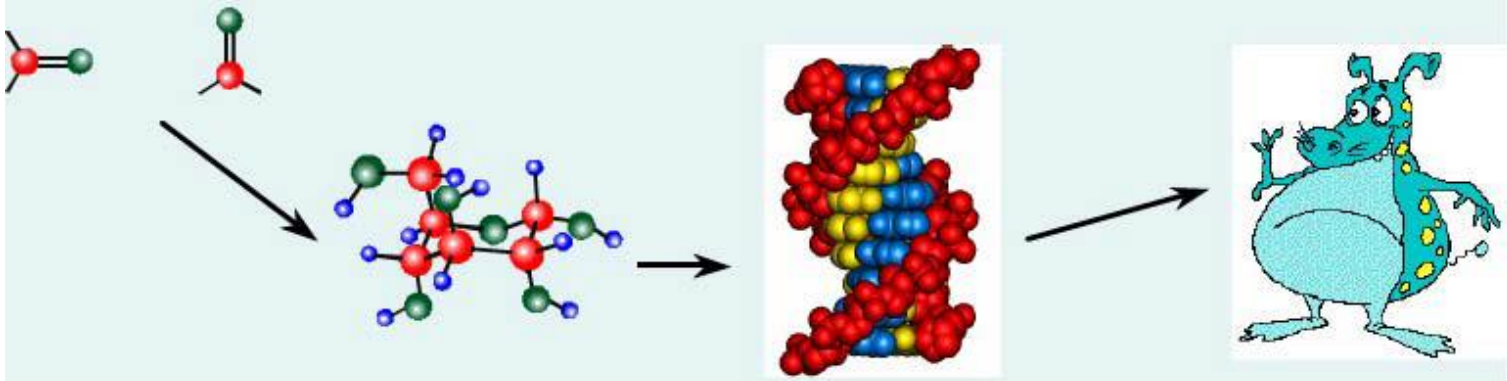
Basic prebiotic chemistry

Prebiotic chemistry is organic chemistry in aqueous solution, under plausible conditions of the primitive terrestrial environment, leading to compounds of biological interest. Elementary prebiotic chemistry uses simple and reactive organic compounds, such HCN, HC3N, HCHO or their oligomers.

Hydrogen cyanide (HCN) Cyanoacetylene(HC3N)

Formaldehyde (HCHO) is the most important carcinogen

致癌物

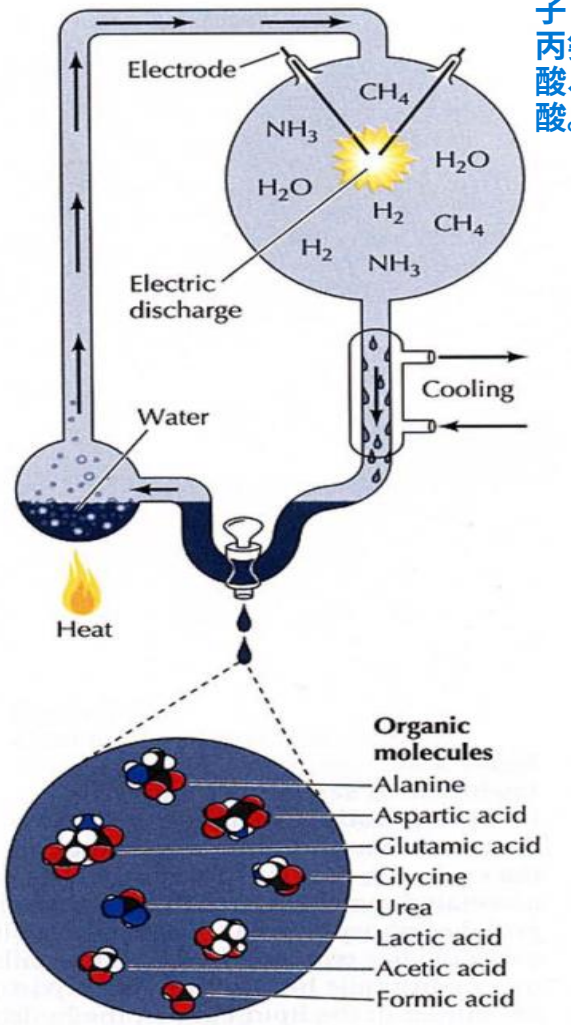


Miller-Urey Experiment (1953)

有机分子的自发形成

Spontaneous formation of organic molecules Water vapor was refluxed through an atmosphere consisting of CH_4, NH_3, and H_2, into which electric sparks were discharged. Analysis of the reaction products revealed the formation of a variety of organic molecules, including the amino acids alanine, aspartic acid, glutamic acid, and glycine.

水蒸气通过由甲烷(CH4)、**氨**(NH3)**和**氢气(H2)组成的大气环境，同时通入电火花放电。对反应产物的分析显示**形成了多种有机分子，包括氨基酸：丙氨酸、天门冬氨酸、谷氨酸和甘氨酸。**



Atmosphere from volcanic outgassing

Problem: The assumed atmospheric composition

The experiments only give large yields of interesting organics (amino acids, nucleic acids, sugars) if the gas is **H-rich** (highly reducing). If the early atmosphere was CO_2+ N_2 (mildly reducing), as many suspect, the yields are tiny.

有机物

强还原性

This would give atmosphere rich in CO_2, N_2, and H_2O. ***Not*** the composition that favors Miller-Urey synthesis.

Prebiotic Chemistry

How to Make Some Critters 生物

Evolutionary Advantage Drives Everything!

Step 1: Rapid chemical synthesis of raw materials in the atmosphere produces amino acids:

$$NH_3 + 2CH_4 + 2H_20 + energy = C_2H_5O_2N + 5H_2$$

glycine

$$5H_2CO = Ribose(food)$$

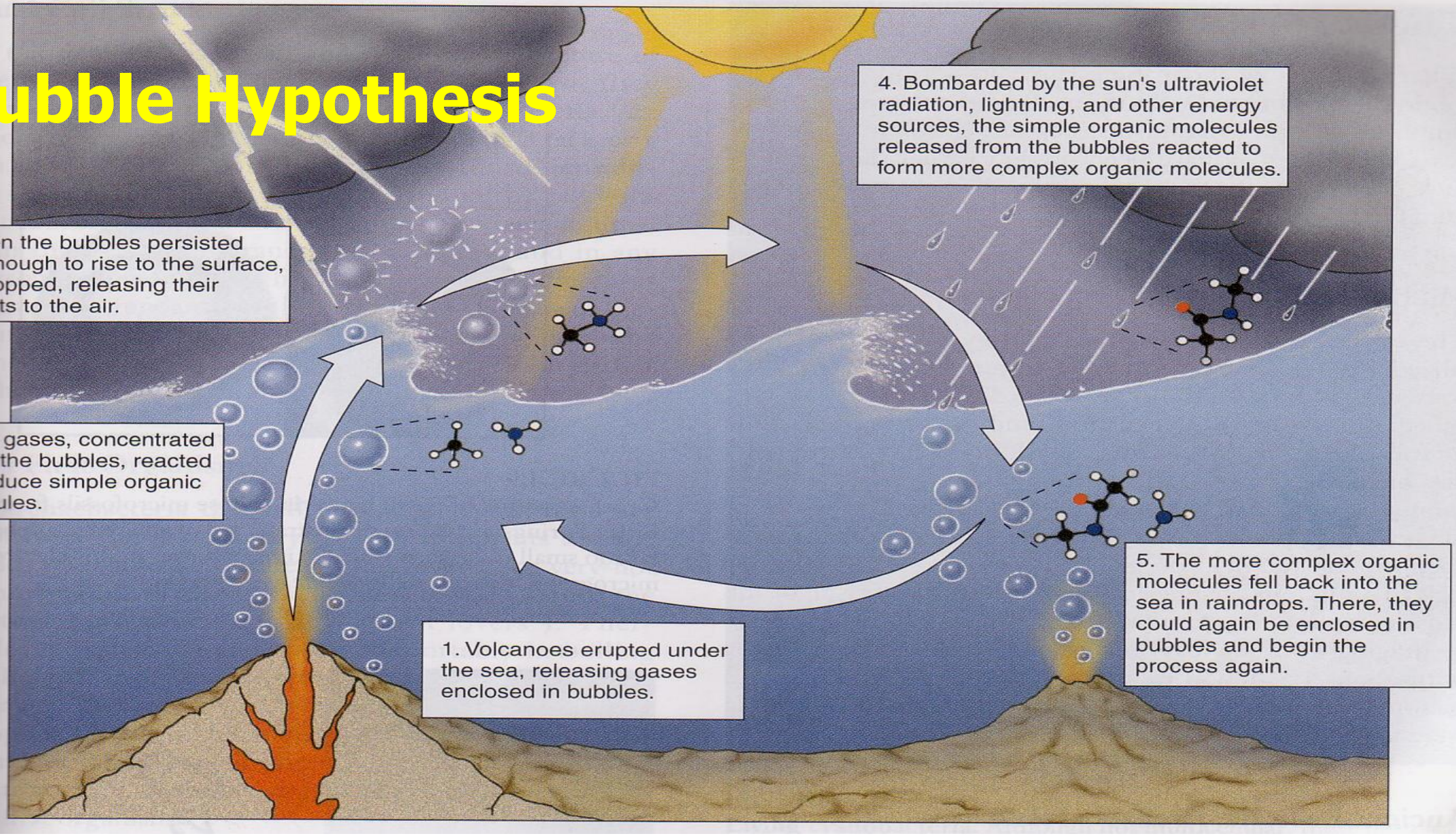
HCHO formaldehyde 甲醛

$$5HCN + UV + NH_3 = Adenine$$

水蒸气凝结

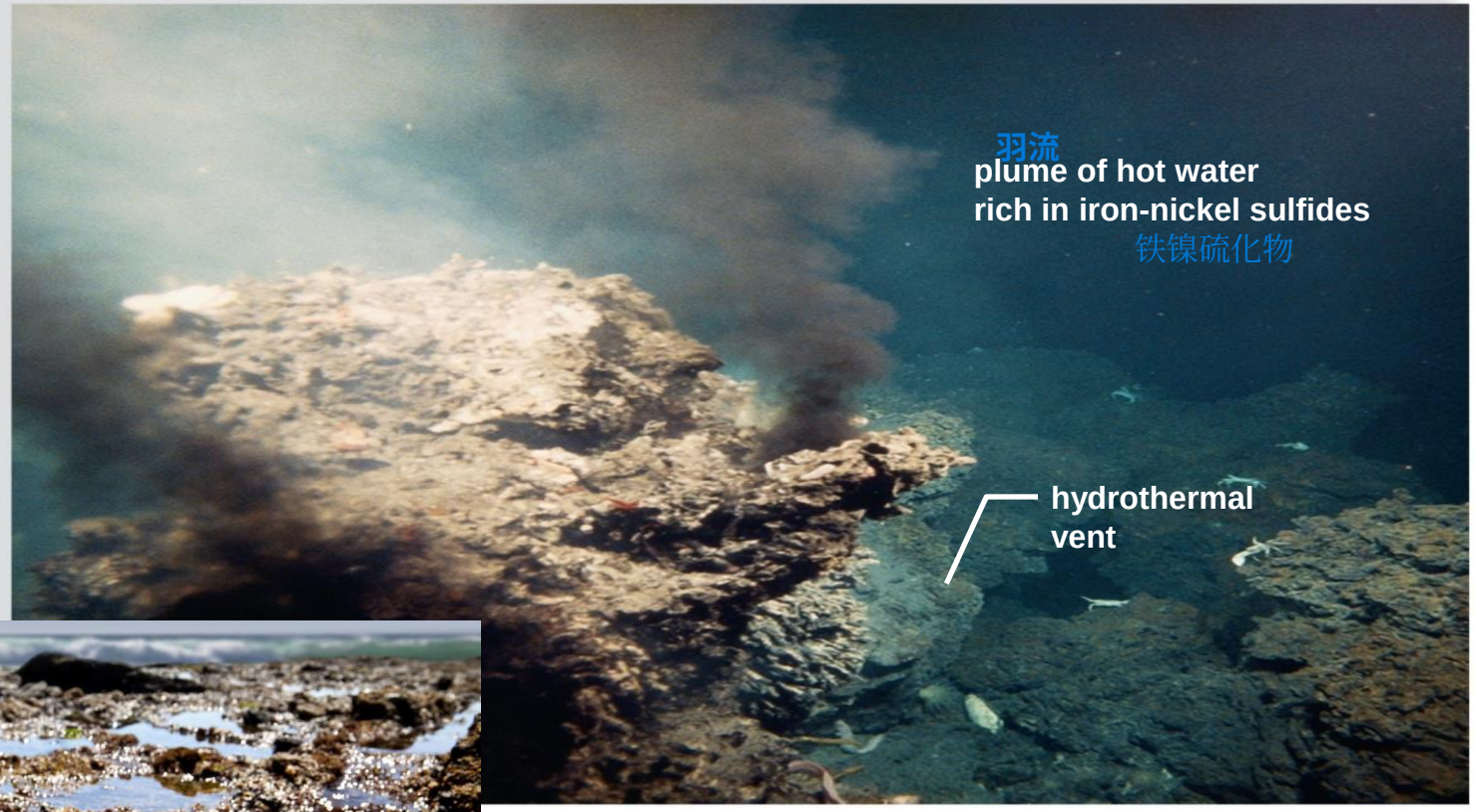
After condensation of water vapor produced the earth's oceans, thus sweeping out the carbon dioxide and locking it up into rocks, our atmosphere was mostly nitrogen.

Bubble Hypothesis



In 1986, geophysicist Louis Lerman proposed that the chemical processes leading to the evolution of life took place within bubbles on the ocean's surface

Evolution of Small Organic Molecules

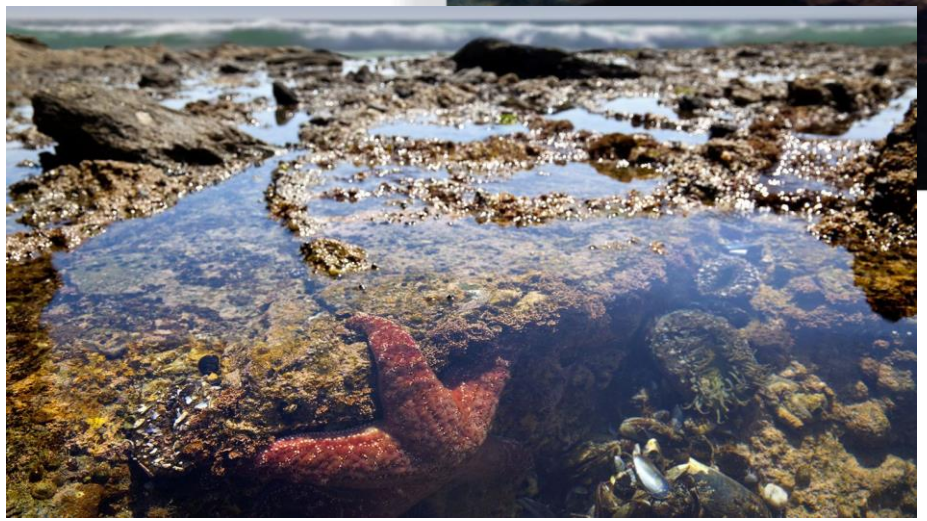


Some scientists hypothesize life began in hydrothermal vents deep in the ocean

深海的热液喷口

Some scientists hypothesize life began in tidalpool

潮池



anaerobic microorganisms

厌氧微生物

CO2+sunlight: photosynthesis

光合作用

Requirements for a life

How to create a life,

合成

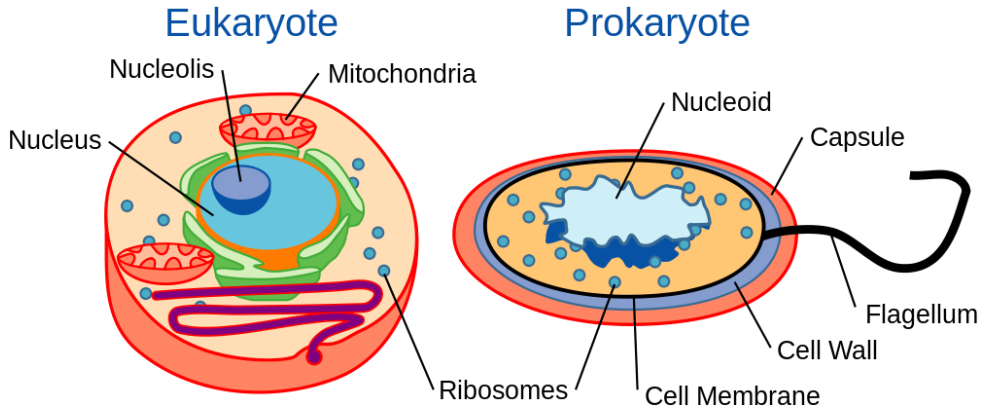
(1)this system must be able to synthesize needed building constituents of living materials from environment (Metabolism),

新陈代谢

(2)The maintenance of a relatively stable internal (local) chemical environment (Cell),

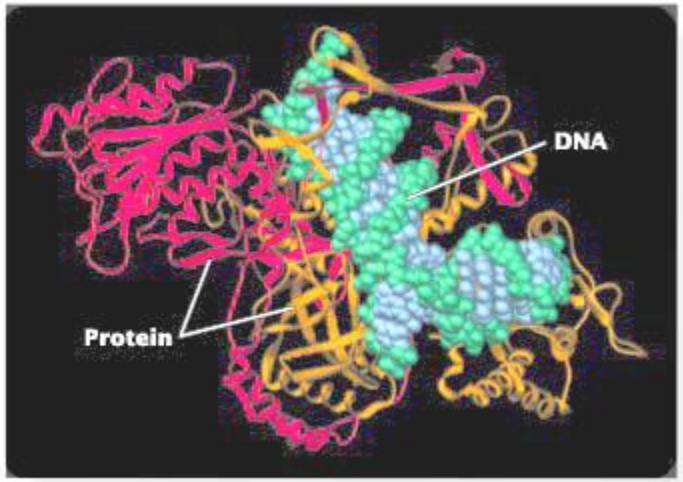
(3)pass on genetic material from one generation to next generation,

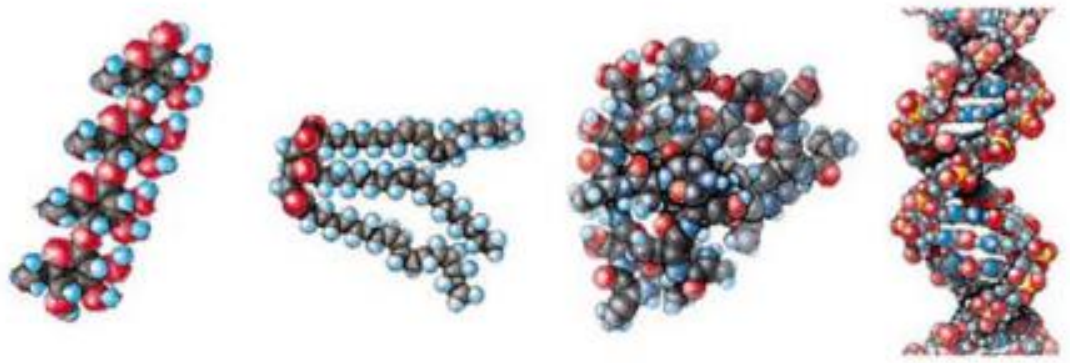
(4)the system can harness energy from environment. All these require some enzymes if you want to build a molecular life form



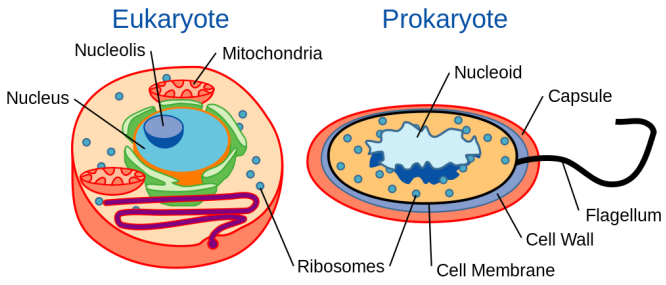
生物聚合物

Four Types of Biopolymers





Carbohydrate (starch) | Lipid (triacylglycerol) | Protein (enzyme) | Nucleic acid (DNA)



DNA-protein system: Too complex for the first life
So which came 1st?

DNA → RNA → Protein

原始

- ***However, only two types seem crucial for primitive biological processes:***
- ***amino acids/proteins and nucleotides/nucleic acids***

氨基酸/蛋白质和核苷酸/核酸

Theories about the origin of life

- RNA-first Hypothesis
  - 核酶 Ribozymes are RNA molecules that can function as both a genetic substrate and an enzyme
  - If RNA evolved first it could function as both genes and enzymes
  - Some viruses have RNA genes
  - 逆转录酶 Reverse transcriptase produces DNA from RNA
  - Suggests a mechanism as to how cells evolved to have DNA genes

- Protein-first Hypothesis
  - Amino acids can form polypeptides when exposed to dry heat 干热条件形成多肽
  - Form microspheres when introduced back into water 微球体
  - Complex enzymatic processes may have been necessary for formation of DNA and RNA 酶促过程
  - Enzymes may have been needed to produce nucleotides and nucleic acids

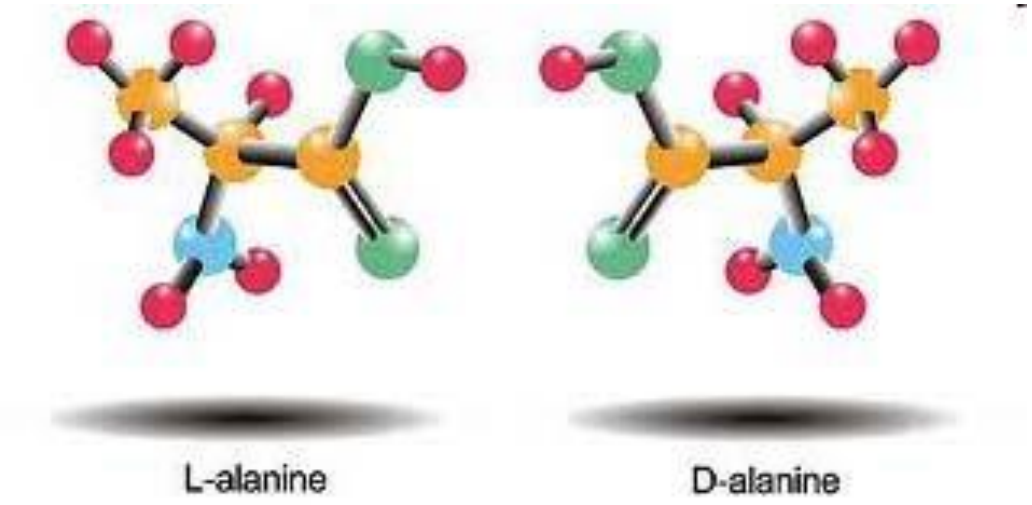
Pre-RNA world

So now we have some amino acids 单体(monomers) loosely mixed in the oceans. Liquid medium is important:

- Ease of transport and Interaction
- Next goal is to combine Monomers into Polymers (peptide chains)

Protects molecules from UV photon disruption

我们认为地球生命选择旋光性纯属偶然，
如果宇宙其他地方存在碳基生命形式，其
化学结构理论上可能具有相反的旋光性。



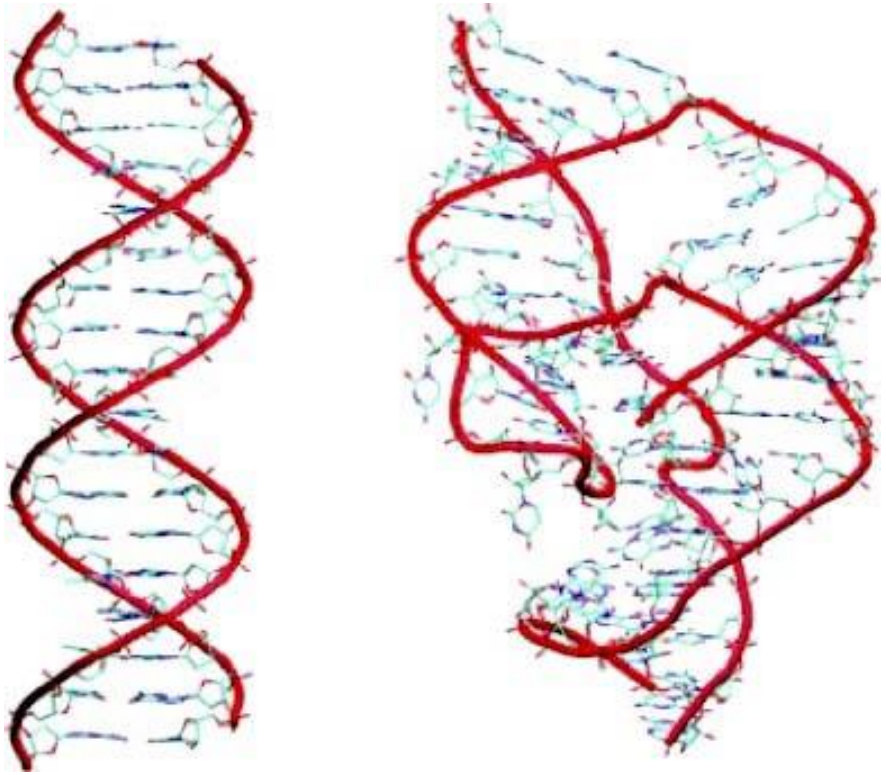
Most amino acids have a mirror image (L and D):

- L and D both found in meteorites
- L only in organisms on the earth

why is D selected against?(*)

(*) We believe that Earth life's "choice" of chirality was purely random, and that if carbon-based life forms exist elsewhere in the universe, their chemistry could theoretically have opposite chirality.

RNA world



- **What came before DNA and proteins?**

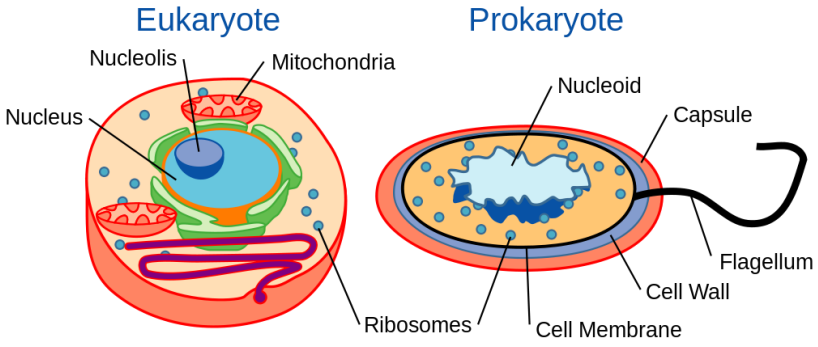
  Almost certainly: RNA
- RNA looks a lot like DNA, but is single stranded. The big difference is that RNA is a molecule that can carry information like DNA, but can also fold itself into complex three-dimensional shapes like proteins,
- **so RNAs can be their own enzymes (proteins).** Because RNA is *ribo*nucleic acid, but can act like an enzyme (protein), these primordial RNAs are called "**ribozymes**" and are the most important candidate for the origin of life.

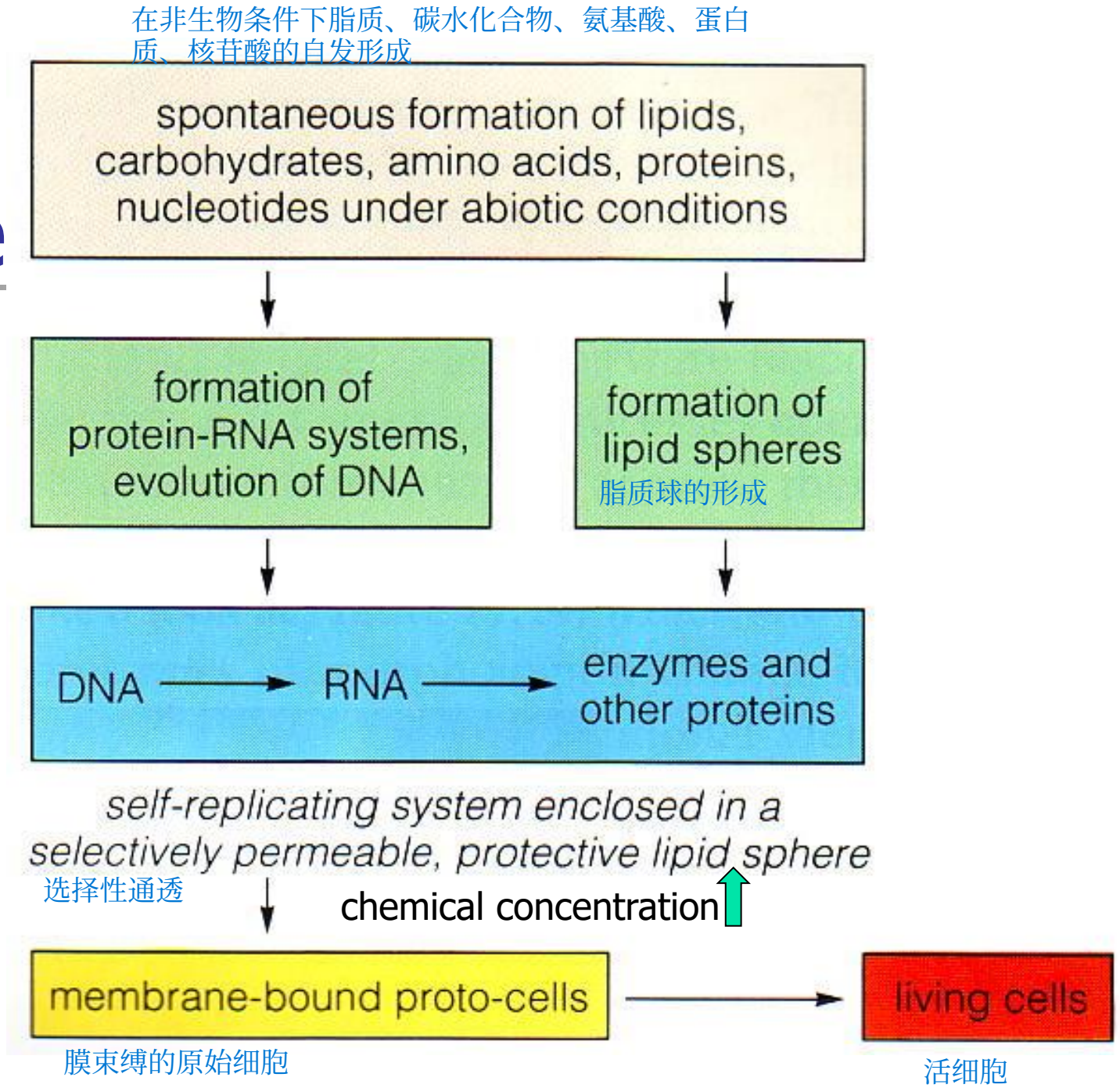
  原始 核酶
- ***When naturally occurring ribozymes were discovered in present-day organisms (including humans), the idea that there was once an "RNA world" became easily the most plausible scenario for the transition to life.***

当在当今的生物（包括人类）体内发现天然存在的核酶时，
曾经存在一个"RNA 世界"的观点便顺理成章地成为
生命起源过渡阶段最合理的设想

Theories about the origin of life

One proposed sequence of events that led to the first **self-replicating** 自我复制系统 **system** and, later, to the first living cells.





How self-replicating RNA could have led to the DNA/protein world

A strand of RNA serves as a template for its own replication.

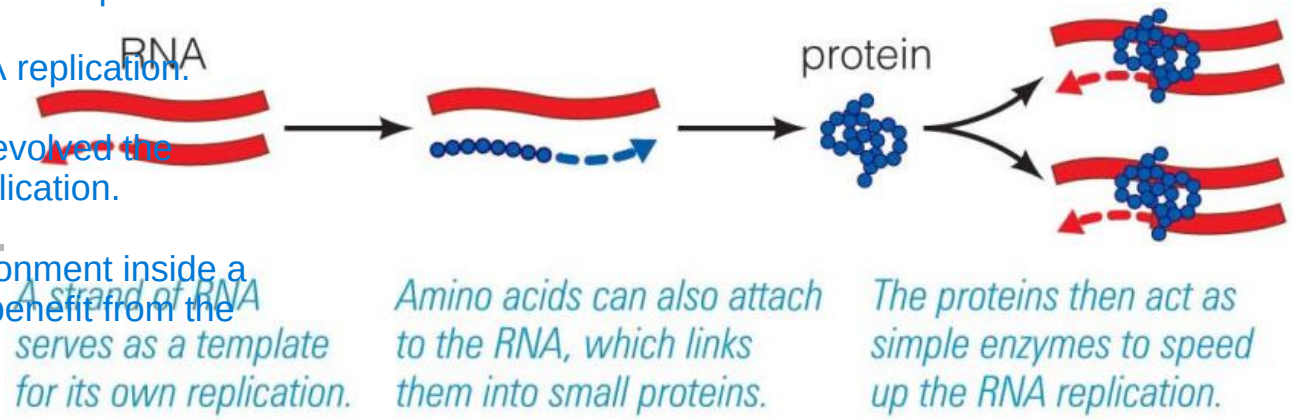
Amino acids can also attach to the RNA, which links them into small proteins.

The proteins then act as simple enzymes to speed up the RNA replication.

This diagram shows a self-replicating RNA molecule that has evolved the capability to produce a primitive enzyme that helps its own replication.

If the RNA and the enzyme are isolated from the outside environment inside a pre-cell, then only the molecules in this particular pre-cell will benefit from the new enzyme, a fact that can speed up molecular evolution.

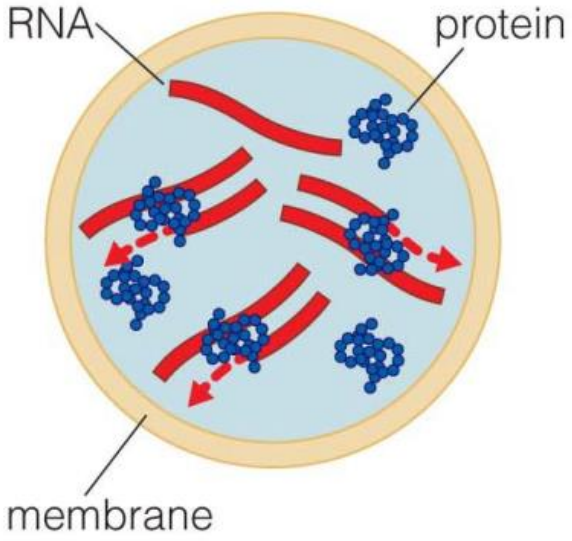
中文翻译：



a This diagram shows a self-replicating RNA molecule that has evolved the capability to produce a primitive enzyme that helps its own replication.

如果RNA和酶被隔离在前细胞内部环境**中，**那么只有这个特定前细胞中的分子才能从新的酶中受益，这一事实可以加速分子进化。

b If the RNA and the enzyme are isolated from the outside environment inside a pre-cell, then only the molecules in this particular pre-cell will benefit from the new enzyme, a fact that can speed up the molecular evolution.



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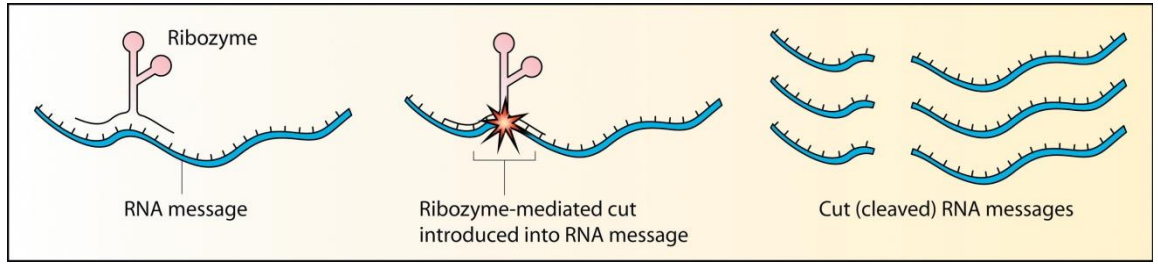
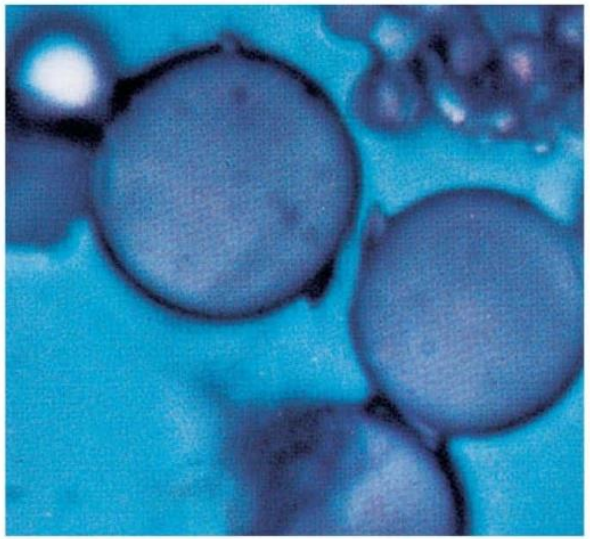


1989 Nobel Prize in Chemistry Cech&Altman

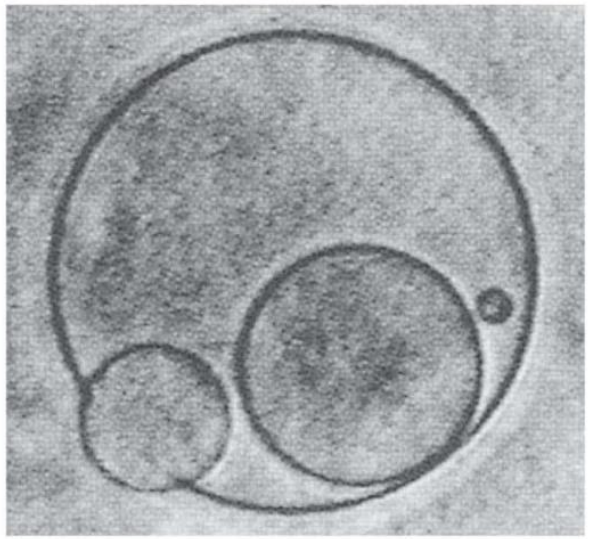
Encapsulation: Prerequisite for RNA world? 封装

益生元膜（囊泡）

Prebiotic membranes (vesicles) are easy



a These microscopic spheres were made by cooling a warm-water solution of amino acids. They are not alive, but they exhibit many lifelike properties.



b These microscopic membranes are made from lipids that, when mixed with water, spontaneously form enclosed droplets.

液滴

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RNA聚合物

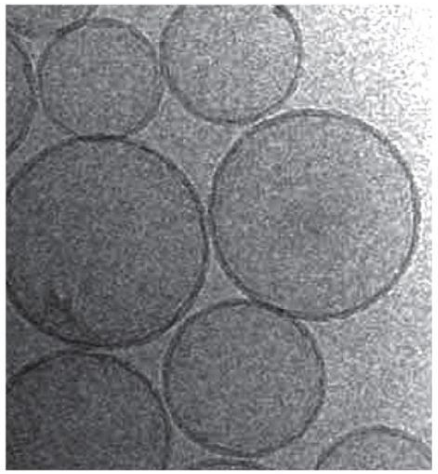
- The production of RNA polymers at fast enough rate is usually considered a problem, but there are many ways to enhance it. One is to confine the reactants to a compartment of some kind; a lipid vesicle, forerunner of today's lipid membranes.

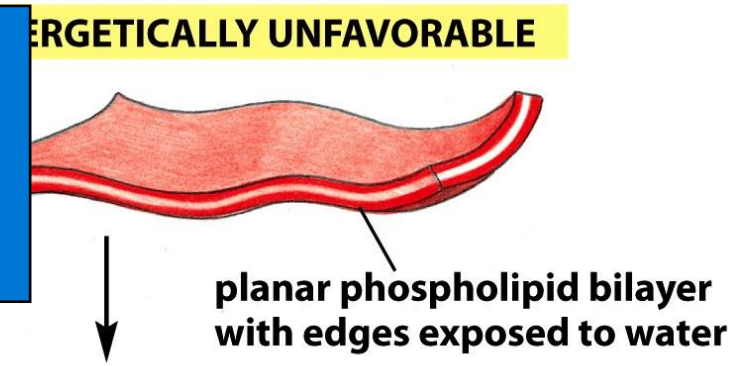
其中一种方法是将反应物限制在某种隔室中；比如脂质囊泡，它是当今脂质膜的前身。

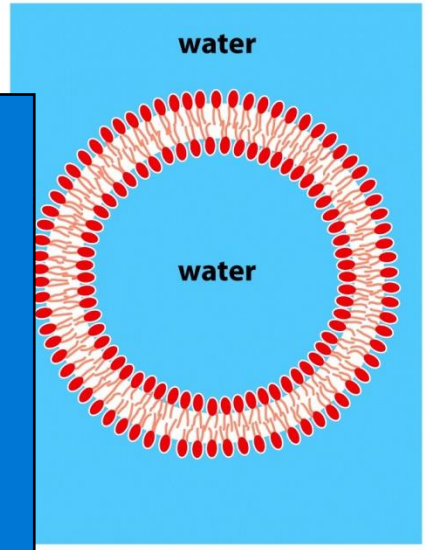
Evolution of the Membrane System

Plasma membrane → protocells

质膜-原细胞







(A)

(B)

Figure 11-12 Essential Cell Biology 3/e (© Garland Science 2010)

Cairns-Smith hypothesis (DNA/Protein)

黏土

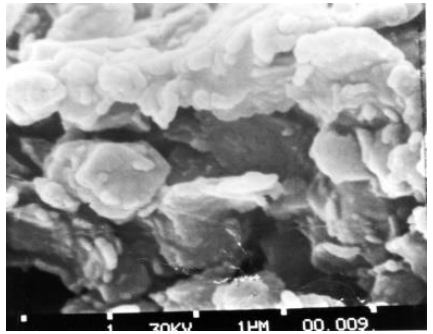
Clay was helpful in polymerization of both proteins and nucleic acids at the same time

May have served as a catalyst and energy source

细粒

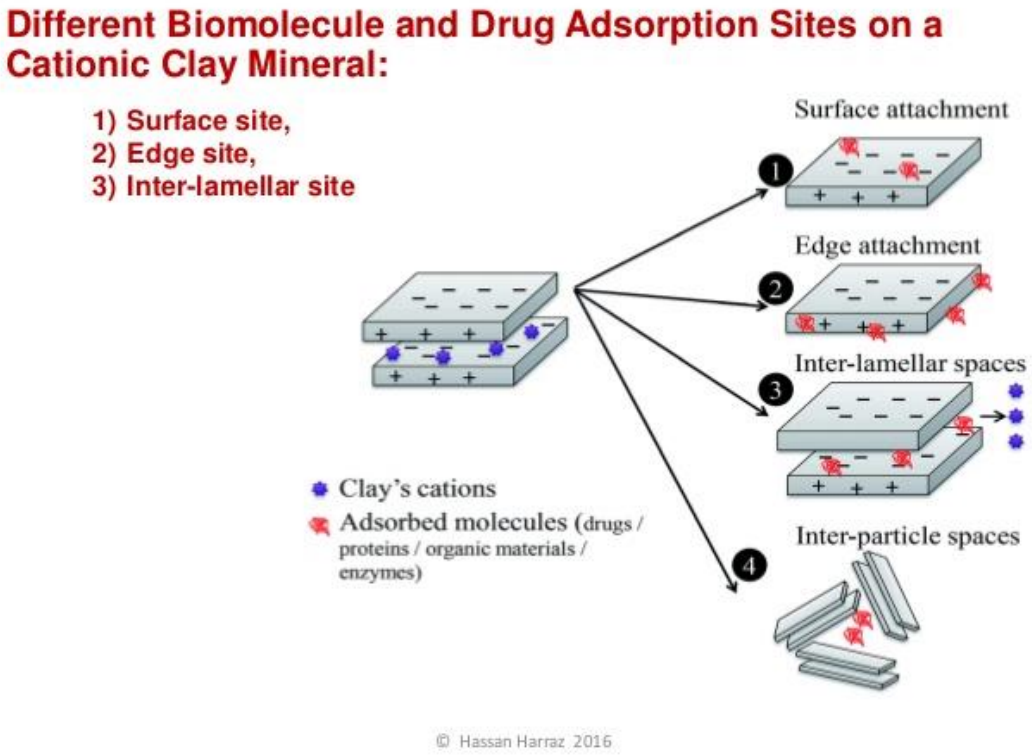
Clay is a type of fine-grained natural soil material

Clay crystal masses of a particular external form may happen to affect their environment in ways that affect their chances of further replication



合成复合物

Formation of synthetic complex (protein synthesis)



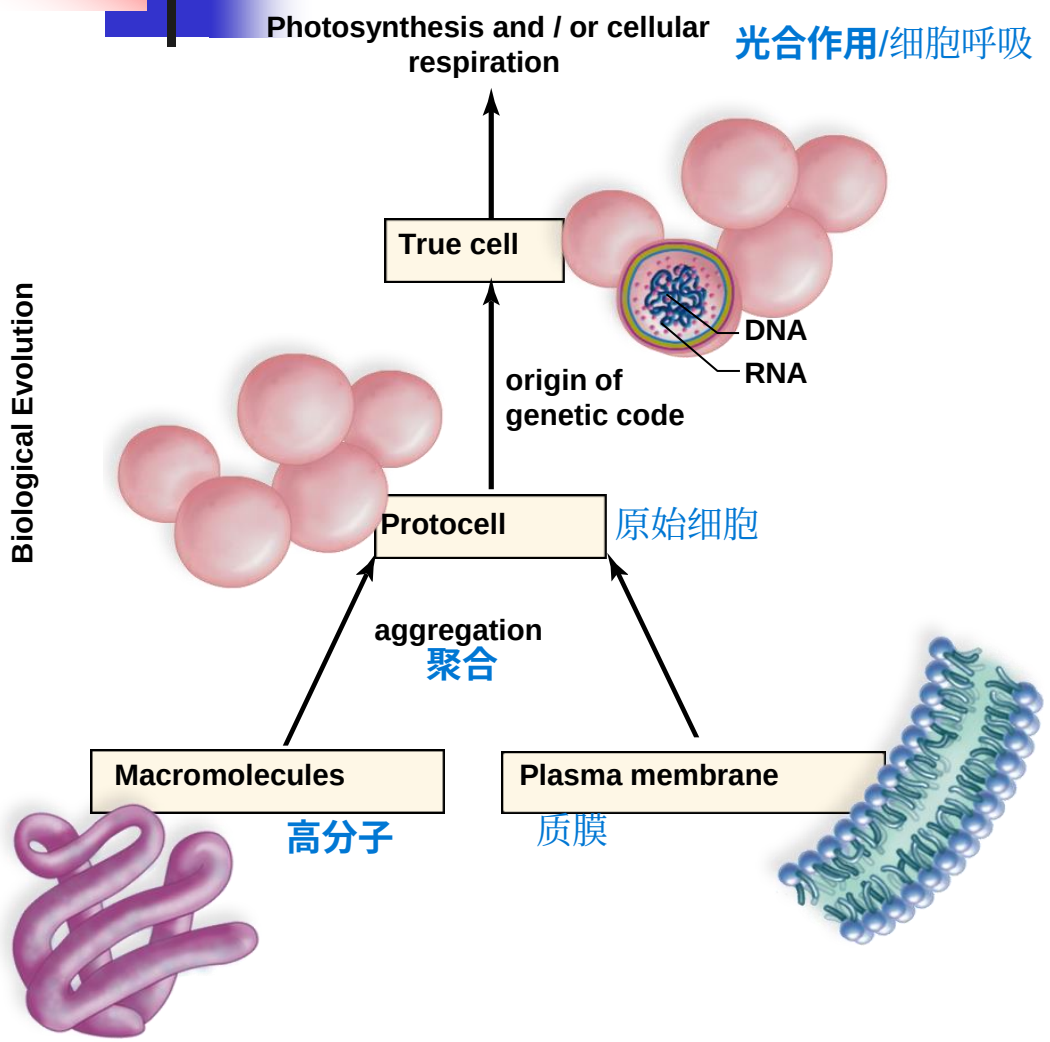
Primordial genetic molecules adsorbed on clay minerals would have been protected against degrading agents present in the environment and would have been in the right conditions to undergo evolutionary processes.(Gene 2005)

吸附在黏土矿物上的原始遗传分子会受到保护，免受环境中降解剂的影响，并处于适宜的条件下进行进化过程。

RNA genes could replicate because **proteins** were already present to catalyze the reactions

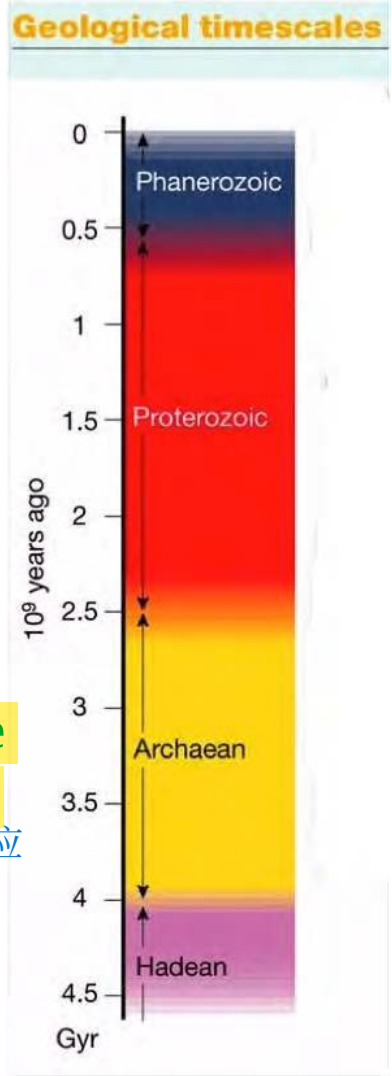
But this supposes that **two** unlikely **spontaneous** 自发 **processes** would occur at once - formation of **RNA** and formation of **protein**

True Cells 真细胞



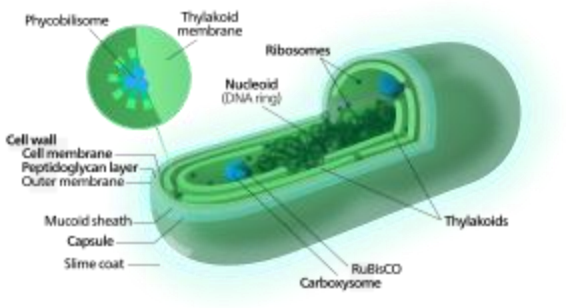
When did life begin?

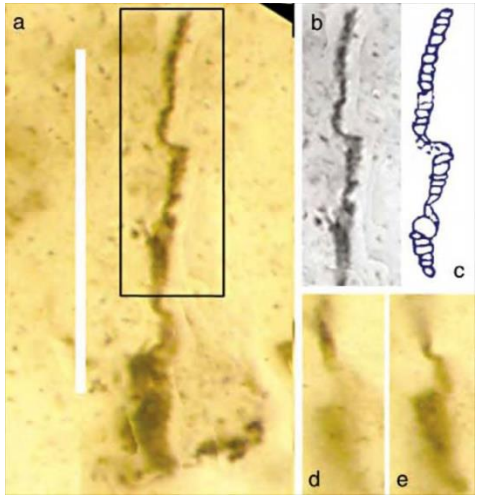
- **Stromatolites:** Bacterial colonies that used Photosynthesis (Cyanobacteria) 蓝细菌 硫化叶状体
- **Microfossils:** Difficult! Controversial…. 微粒化石
- **Isotope ratios**: carbon-12 to carbon-13 abundance is affected by metabolism in living things. 同位素比率
- **Best estimate: 3.5 to 4.0 Byr ago**
- When organisms ingest carbon, they preferentially use 12C over 13C. (14C is radioactive, and thus won't remain over a long time period.) **Carbon with a high ratio of 12C compared to 13C is therefore an indicator of living processes (diffusion /reaction rates of 12CO2 >13CO2).** 扩散/反应 Carbon enriched in 12C has been identified in rocks from Greenland dated at 3.85 billions of years ago. This is the earliest evidence for life on Earth.



Establishing biological nature of fossils: stromatolites 叠层石

化石



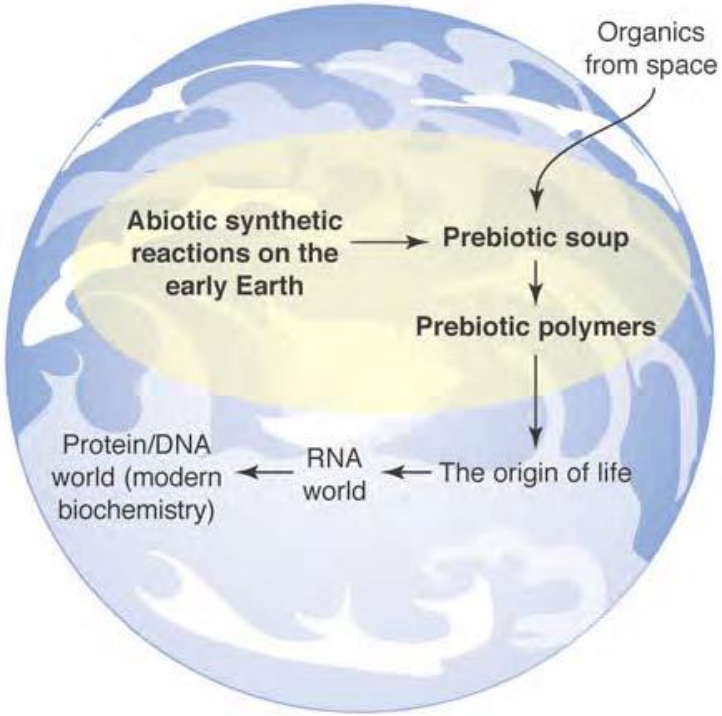


Science 8 March 2002 : "Earliest Signs of Life Just Oddly Shaped Crud?"

前寒武纪岩石

- **stromatolites** found in Precambrian rocks as the earliest known fossils, and still being formed in lagoons in Australasia.
- The Earliest Trace of Life? This fossil from Western Australia is 3.5 billion years old and shows carbon traces that indicate life. Its form is similar to that of modern filamentous cyanobacteria (inset). 现代丝状蓝细菌
- Oldest stromatolites are about 3 Byr, but photosynthesis is so complex that it could not have been available near the beginning of life. 光合作用如此复杂，**不可能在生命起源之初就存在。**
- Photosynthesis from algae in the ocean produced most oxygen in earth atmosphere 海洋中藻类的光合作用产生了地球大气中大部分的氧气。

Where did life begin?



- **Land?**

  杀菌冲击

  Problem: No protection from intense UV, or from sterilizing impacts.

  Additional reason for excluding origin on land:

  Hard to imagine life not in an aqueous solution. **不在水溶液**

- **Ocean?**

  聚合

  How to concentrate the molecules so they polymerize in a reasonable time? One possibility: ***Encapsulation of molecules in cell-like membrane.*** 将分子封装在类似细胞的膜中

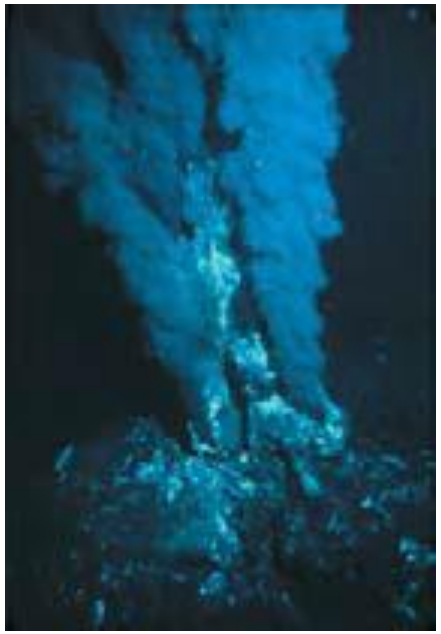
- **In tidepools or lagoons?** **潮池or泻湖**

  Evaporation concentrates monomers,

  but unfortunately exposes to UV. 蒸发可使单体浓缩

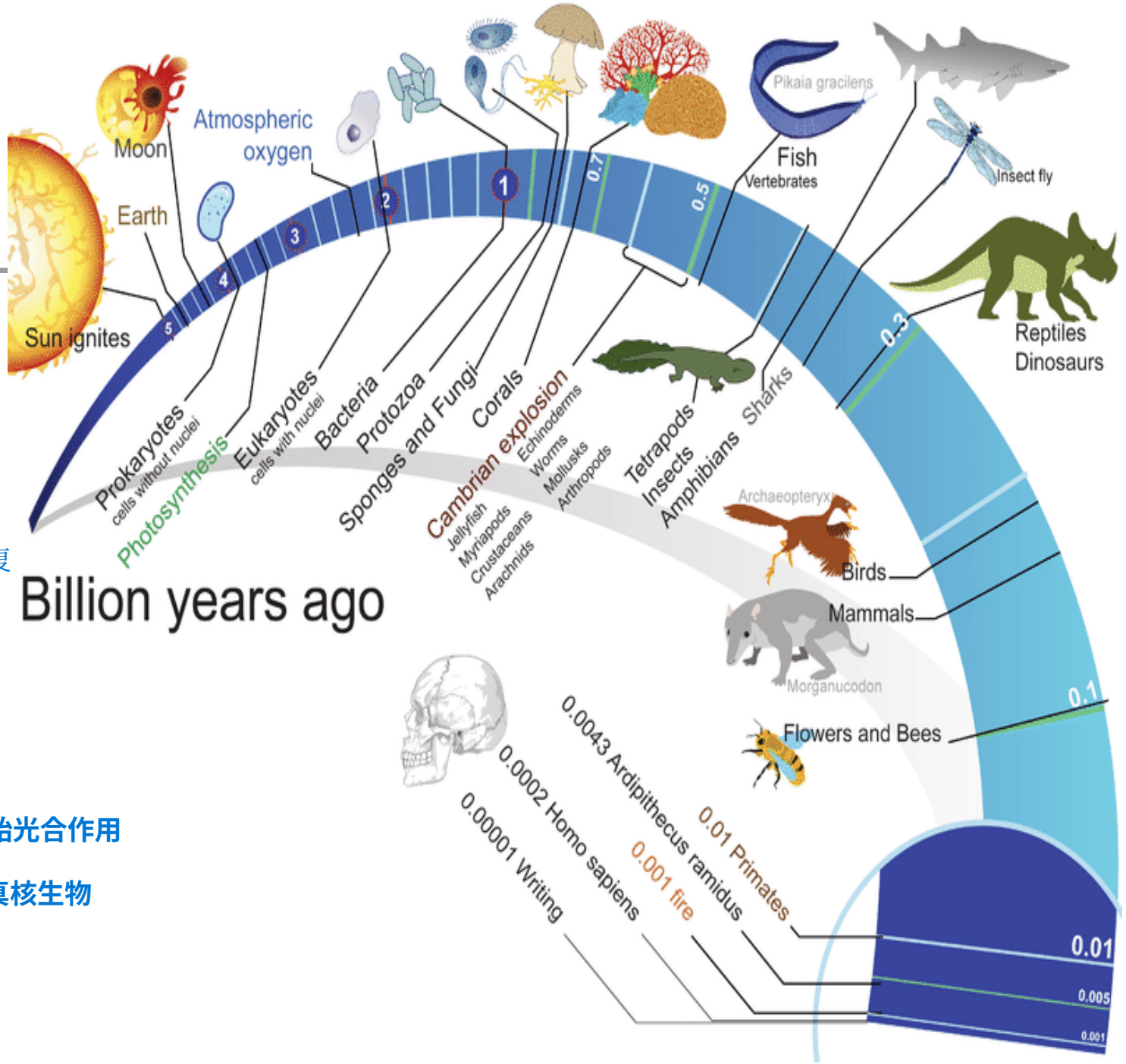
- **Hydrothermal deep-sea vents?**

  A present-day favorite



Life on Earth

- Earth forms over a time of 50Myr more than 4.5 Byrs ago
- Pre-biotic chemistry somehow leads to first replicating macromolecules (maybe RNA) 前生命化学以某种方式导致了第一批能够自我复制的大分子（可能是核糖核酸）的出现
- RNA leads to DNA and first life form(s)
- Best estimate: life on Earth is between 3.5 and 4 Byrs old! 35亿年-40亿年
- For >2 Byrs we have simple prokaryotes, they start photosynthesis 简单的原核生物，开始光合作用
- 1.5 Byrs ago Eukaryotes evolve 15亿年，真核生物
- ~600 Myrs ago complex, multi-cellular life evolves 6亿年，复杂多细胞生命



Lucy, the mother of mankind

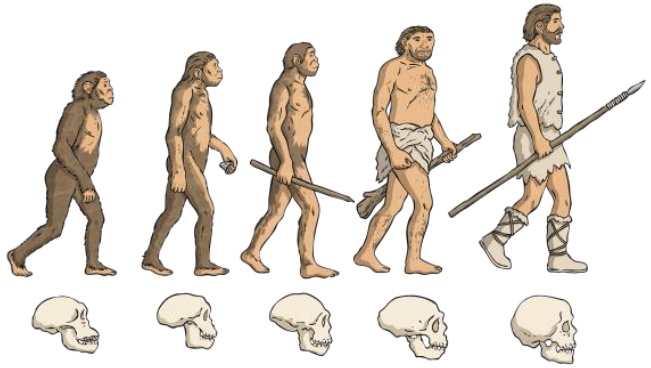
Perhaps the world's most famous early human ancestor, the 3.2-million-year-old ape "Lucy" was the first Australopithecus afarensis skeleton ever found, though her remains are only about 40 percent complete (photo of Lucy's bones)

对于“非洲起源说”存在广泛共识，该假说认为
现代人类约在 20 **万年前起源于非洲**，并在约 10 万年前遍布整个非洲大陆。

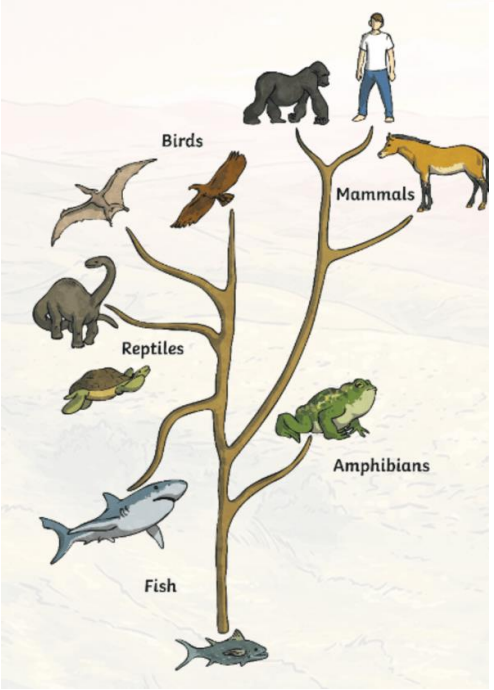
There is broad consensus for the 'Out-of-Africa' hypothesis that states that modern humans arose ~200 000 years ago in Africa and spread throughout the continent ~100 000 years ago.

在人类遗传学中，线粒体夏娃（更专业的术语是线粒体最近共同祖先（MRCA））。即使是亲缘关系很近的人类群体，**其** MRCA 的估计时间也远超 5000 **年前**。

In human genetics, the Mitochondrial Eve (more technically known as the Mitochondrial-Most Recent Common Ancestor (MRCA). Estimates of the MRCA of even closely related human populations have been much more than 5000 years ago



Lucy (2014 movie)



The Cambrian Explosion 寒武纪大爆发

- 530 Myrs ago, fossil record of animals and other complex organisms "explodes"

多样化

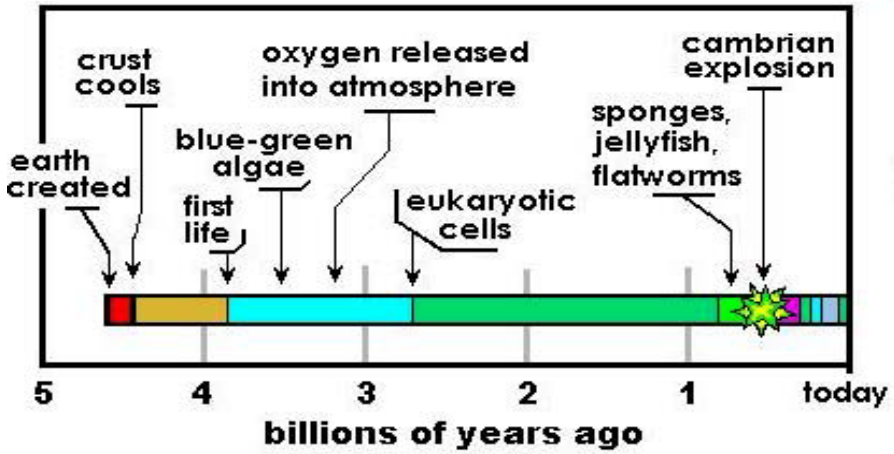
- Major diversification of life on Earth
- Explosion took many millions of years (organisms before 580 Myrs were much simpler)

Several hypotheses:

– Increased oxygen levels (more energy)?

– Earth was recovering from a Snowball event (several frozen period)?

– Evolution of eyes (sensing far distance in 3D space) ?

– It wasn't an explosion at all!

*Nature***530,** 268–270(18 February 2016)